Evidence-Bounded 3D Damage Detection from Sparse Views and a Healthy CAD Model

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The problem

A dark spot in a photo of a turbine blade could be any of these:



DENT real 3D damage | **STAIN** only the colour changed | **LIGHT** shadow or brighter lamp | **CAMERA** camera slightly off

In a photo they can look almost the same: a 0.1 mm dent changes the image only slightly, while a 1° camera error changes thousands of pixels. Current methods either mark unusual pixels in 2D or build a 3D copy of the object; **none tells these causes apart while measuring the 3D damage, and none says when the photos are not enough.**

Our idea

Input: healthy CAD model + 2–7 calibrated photos.
Output: the damage in 3D (location and depth in mm), *or* "not enough evidence".

The method only reports damage that the photos can actually prove.



Method

1. **Explain without damage:** fit camera position, lighting and a possible stain so that a rendering of the healthy CAD matches the photos as well as possible.
2. **Find suspicious regions:** a fast statistical check over the whole blade finds where a dent would explain the remaining difference best.
3. **Fit a dent:** at those places, fit a smooth 3D dent (position, depth, size, shape) by differentiable rendering: render, compare with the photos, adjust.
4. **Evidence test:** *evidence* = mismatch without damage − mismatch with the dent. Report the dent only if the evidence exceeds a threshold set on healthy and stained blades (so false alarms stay near 5%); otherwise abstain.

Key results (computer-rendered blades; sealed test: 240 cases, 30 unseen blades)

Method	AUROC	Dents found	Located	False alarms
Ours	**0.82**	**54%**	**54%**	10%
MLP field*	0.74	16%	11%	6%
2D detector	0.72	24%	16%	4%
Free-form field*	0.63	14%	11%	4%

*Our re-implementations of the CADSplat idea. Each method's threshold is set on healthy and stained blades. At exactly equal false alarms (5%): ours 51%, others at most 19%.

The evidence test matters:
- false alarms 32% → 10%
- hidden dents wrongly "found" 55% → 15%

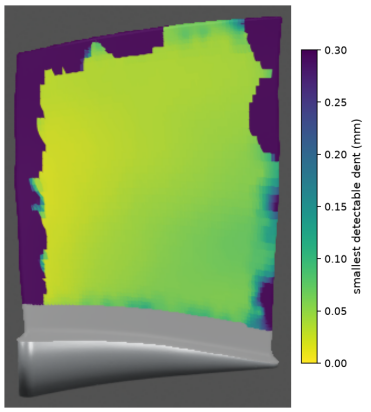
Depth: 12 of 13 dents deeper than 0.2 mm found; most dents below 0.1 mm are declined instead of guessed.

Transfer to new blade designs

Held-out industrial blades 3–6	Frozen v1	v1.1 (fixed)
False alarms	51%	**4%**
AUROC	0.72	**0.95**
Hidden dents correctly declined	50%	**98%**

Six industrial blade CAD designs. Two causes of the failure were found and fixed on blades 1–2 without retraining; blades 3–6 were never used for development.

Detectability prediction



Before searching, the method predicts how much evidence a dent at each point would produce, i.e. which dents the photos can reveal. Map: bright = tiny dents detectable, dark = only deep dents, grey = not inspected.

- **Per dent, the prediction matches reality:** it separates found from missed dents with AUROC 0.90–0.95 on two sealed tests.
- **The per-point map is an upper bound:** it is optimistic for dents shallower than 0.2 mm, so we do not use it as a guaranteed limit.

Identifiability finding

For small, sharp dents seen in two photos, very different depths can produce almost the same pixels: one 0.64 mm dent was found in the right place but measured at 0.06 mm. On new blade designs, dents are measured at about 60% of their true depth; four views instead of two find more dents (45 vs. 36 of 60) but do not improve depth. **Finding *where* damage is works; measuring *how deep* it is needs more information**, most likely low-angle (grazing) lighting.

This agrees with feedback from a PIAD author, who pointed out that geometry, illumination and reflectance cannot be reliably separated without additional constraints, and suggested studying when a method should abstain as a research contribution in itself.

Contributions

1. **Evidence-bounded detection:** a method that reports 3D damage only when the photos support it, with false alarms controlled by calibration (textbook statistical tests give 32% and 96% false alarms instead of the intended 1%).
2. **Detectability prediction:** predicts which dents the photos can reveal (AUROC 0.90–0.95).
3. **Identifiability study:** shows that location is reliable across designs while depth is not, and why.
4. **Transfer fix (v1.1):** false alarms on unseen blade designs from 51% to 4%, without retraining.
5. **Beyond smooth dents:** without any change, it also locates rounded dents on the blade edge (72%) and long shallow scratch-like grooves (66%) on new designs; a standard SIFT detector finds 16–25%. Realistic sharp nicks and hairline cracks are being tested next.

Next step

Validate on real blades: 3D-print industrial blades, add dents, nicks, cracks and stains, 3D-scan them for the true shape, photograph them from 2–7 views under normal and grazing light, and compare.

References

[1] PAD: pose-agnostic anomaly detection dataset (NeurIPS 2023). [2] PIAD: pose and illumination agnostic anomaly detection (CVPR 2025). [3] CADSplat: sparse-view Gaussian splatting aided by CAD models (arXiv 2026). [4] Looking 3D: anomaly detection with 2D-3D alignment (CVPR 2024). [5] Real-IAD D3: a real-world 2D/pseudo-3D/3D dataset for industrial anomaly detection (CVPR 2025). [6] nvdiffrast: modular primitives for differentiable rendering (2020).